VF Asia Footwear Laboratory Testing Manual

Test Method for Bursting Strength	**Test Method #: VF- B02** **REF #: TM-R- 025** **Page 1 of 1**
Date Issued : May,2019 **Revision #: 1** **Reason for revision: Rectify the unit of measure from kg /cm^2 to kgf /cm^2**	**Created by: Nila Babia Erazo**

PURPOSE AND SCOPE: To measure the pressure required to rupture a material by distending it with a force which is applied at a right angle to the surface of the material.

REFERENCES: ASTM D3786

EQUIPMENT AND MATERIAL:

1. Bursting strength tester, meeting the requirements of ASTM D 3786.
2. Leather, fabrics, cartons

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory environment at 23.0±2.0 ˚C Temperature and 55 ±10 % Relative Humidity.
2. Die or hand cut a 10cm *10cm test sample from the test material. Samples must be taken from different areas of the test material, and at least 10cm from any edge.

Note: With fabrics, the test samples should be taken such that no two samples contain the same warp or fill yarns.

REPETITONS: Test three (3) samples per material.

TEST PROCEDURE:

Test should be carried out at standard laboratory conditions @ 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.

1. Return the pointer on the pressure gauge to zero.
2. Place the test sample between the compression discs of the bursting strength tester.
3. Pull the test material taut and clamp the test material firmly between the compression discs of the bursting strength tester. Avoid excessive clamping pressure as this can damage the machine. For machines with clamping pressure gauge, a clamping pressure of 5-7kgf /cm2 is recommended.)
4. Inflate the diaphragm until the material ruptures. Return the inflation handle to neutral immediately after the material ruptures to avoid machine damage.

Note: Do not exceed a pressure of 50kgf /cm2. If a test sample has not burst at 40kgf /cm^{2,} discontinue the test and record the result as 50$^+$ kgf /cm^2.

5. Record the maximum pressure required to rupture the material to the nearest 0.5kgf /cm2.

REPORT: Report the average of the three readings to the nearest 0.5kgf /cm2.